SSC MTS 2024 question paper with answers

17 October 2024 · Shift 1

77 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 13 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 15 questions

Q.2

Pure ghee costs ₹100 per kg. After adulterating it with vegetable oil costing ₹50 per kg, a shopkeeper sells the mixture at the rate of ₹96 per kg, thereby making a profit of 20%. In what ratio does he mix the two?

(A) 3 : 1
(B) 1 : 2
(C) 4 : 1
(D) 3 : 2

Q.3

A cone of radius 16 cm and height 12 cm is cut, parallel to the base, at some height closer to the top. If the ratio of the heights of the solids is 1:3, then the ratio of the total surface areas of the two solids formed is approximately given by:

(A) 1 : 8
(B) 7 : 180
(C) 9 : 143
(D) 2 : 15

Q.4

A student's overall grade is determined by their scores in three subjects: Mathematics, English, and Science, with weights of 40%, 30%, and 30%, respectively. If the student scores 80, 90, and 75 in these subjects, respectively, the overall grade of the student is:

(A) 82.5
(B) 82
(C) 81.5
(D) 81

Q.5

A man’s average driving speed for 9 hours is 88 km/hr. During the first five hours of his journey, he drove at 74 km/hr, and last two hours he drove at 82 km/hr. What was his average speed (in km/hr) for the sixth and the seventh hour?

(A) 120
(B) 101
(C) 92
(D) 129

Q.6

By selling an article for ₹20, a dealer makes a profit of 10%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 45%?

(A) 7.36
(B) 8.36
(C) 6.36
(D) 9.36

Q.7

A shopkeeper offers two schemes, the first being a single discount of 30%, and the second being two successive discounts of 25% and 5%. What is the difference (in ₹) in the discounts of both the schemes if the customer did the shopping of ₹5,000 each in both the schemes?

(A) 67.50
(B) 62.50
(C) 65
(D) 70

Q.8

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 8% per year rate of interest for 2 years.

(A) 440
(B) 400
(C) 360
(D) 340

Q.9

The length and the breadth of the floor of a rectangular hall are 98 feet and 70 feet, respectively. If all persons in the hall are allotted equal square spaces as workstations, what will be the area of the largest such workstation that can be allotted to any user, assuming no space in the hall is left unoccupied?

(A) 256 sq. ft
(B) 196 sq. ft
(C) 100 sq. ft
(D) 144 sq. ft

Q.10

A policeman saw a thief at a distance of 800m. When the policeman started chasing the thief, the thief started running. If the ratio of the speeds of the policeman to the thief is 3:2, then the distance travelled by the policeman to catch the thief is

(A) 2.4 km
(B) 2 km
(C) 3.2 km
(D) 1.6 km

Q.12

Which of the following options is equal to the radius of the new sphere, if three spheres of radii 5cm, 3cm and 4cm are melted into form a new sphere?

(A) 5cm
(B) 6.5cm
(C) 6cm
(D) 8cm

Q.14

A batsman scored the following runs in the last 5 innings. Find his score in the 4th innings if the average score is 43. Innings Score (in runs): 1st - 92; 2nd - 53; 3rd - 35; 4th - X; 5th - 27

(A) 4
(B) 12
(C) 10
(D) 8

Q.15

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 72 kg and 37 kg is:

(A) 49.8
(B) 51.8
(C) 52.8
(D) 50.8

Q.16

The average weight of 20 boys is 60 kg. Together with 30 girls, the average weight of all boys and girls is 54 kg. What is the average weight of 30 girls in kg?

(A) 50
(B) 52
(C) 54
(D) 55

Q.18

The ratio of prices of coffee and tea is 5 : 7 and the ratio of their quantities consumed by a family is 7 : 9. Find the ratio of expenditures on coffee and tea.

(A) 7 : 5
(B) 7 : 9
(C) 9 : 5
(D) 5 : 9

Q.19

Find the simple interest (in ₹) on ₹2000 as a sum borrowed at 6% per year rate of interest for 5 years.

(A) 640
(B) 560
(C) 540
(D) 600

Reasoning Ability and Problem Solving 15 questions

Q.1

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts,decide which of the given conclusions logically follow(s) from the statements. Statements: All parrots are eagles. Some parrots are ostriches. Conclusions: (I) Some eagles are parrots. (II) Some eagles are ostriches.

(A) Only conclusion I follows
(B) Both conclusions I and II follow
(C) Only conclusion II follows
(D) Neither conclusion I nor II follows

Q.2

Which two numbers should be interchanged to make the given equation correct? (24 ÷ 6) + (18 × 4) – 54 + 42 = 99 (Note: Interchange should be done of entire number and not individual digits of a given number)

(A) 4 and 6
(B) 18 and 24
(C) 24 and 54
(D) 54 and 42

Q.5

A, B, C, D, E, F and G are sitting around a round table facing the centre (but not necessarily in the same order). C is sitting to the immediate left of A and immediate right of E. E is sitting to the immediate right of B. D is sitting to the immediate left of B and immediate right of F. A is sitting to the immediate left of G. Who is sitting to the immediate right of G?

(A) E
(B) A
(C) F
(D) D

Q.6

In a certain code language, ‘TURF’ is coded as ‘5691’ and ‘RAFT’ is coded as ‘6935’. What is the code for ‘A’ in the given code language?

(A) 3
(B) 5
(C) 9
(D) 6

Q.9

NUQM is related to PWSO in a certain way based on the English alphabetical order. In the same way, JQMI is related to LSOK. To which of the following is QXTP related, following the same logic?

(A) ZSVR
(B) ZSRV
(C) SZVR
(D) SZRV

Q.10

What should come in place of the question mark (?) in the given series based on the English alphabetical order ? BDG, HJM, NPS, TVY, ?

(A) ZBE
(B) ACF
(C) YAD
(D) XZC

Q.11

What will come in the place of '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 5 + 3 – 10 × 2 ÷ 2 = ?

(A) 13
(B) 17
(C) 12
(D) 15

Q.12

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (8, 64, 2) (12, 12, 1)

(A) (10, 3426, 7)
(B) (6, 214, 6)
(C) (4, 256, 4)
(D) (8, 722, 9)

Q.13

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order).Only two people sit between E and F when counted from the right of E. Only two people sit between F and C. Only three people sit between E and B. D sits to the immediate left of A. How many people sit between A and G when counted from the left of A?

(A) 3
(B) 4
(C) 2
(D) 1

Q.15

What should come in place of the question mark (?) in the given series? 115 117 123 125 131 ?

(A) 133
(B) 132
(C) 135
(D) 134

Q.16

In a certain language, 'INCH' is coded as '9631', and 'THIN' is coded as '3269'. What is the code for 'T' in 'THIN'?

(A) 6
(B) 9
(C) 3
(D) 2

Q.17

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some ants are flies. All flies are bees. Conclusions: (I): No ant is a bee. (II): Some ants are bees.

(A) Neither conclusion (I) nor (II) follows.
(B) Only conclusion (II) follows.
(C) Both conclusions (I) and (II) follow.
(D) Only conclusion (I) follows.

Q.18

ODGM is related to NBFK in a certain way based on the English alphabetical order. In the same way, TAYP is related to SYXN. To which of the following is RJFB related, following the same logic?

(A) PFCC
(B) QHEZ
(C) RHDC
(D) PGDD

Q.19

12 is related to 81 following a certain logic. Following the same logic, 24 is related to 153. To which ofthe following is 19 related, following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 125
(B) 123
(C) 131
(D) 129

Q.20

In a certain code language,'legs hands ears' is coded as 'du ph bq','hands eyes hearts' is coded as 'ak fd du','ears hearts arms' is coded as 'fd bq dy'. What is the code for'eyes' in that language?(Note: All the codes are two-letter coded only.)

(A) du
(B) ph
(C) ak
(D) fd

General Awareness 22 questions

Q.1

It is observed that a child's heart rate is 135 times per minute. The beat frequency of heart is ____.

(A) 1.12 Hz
(B) 8100 Hz
(C) 2.25 Hz
(D) 135 Hz

Q.3

Who first postulated that cells are generated from pre-existing cells?

(A) Rudolf Virchow
(B) Jennifer Doudna
(C) Carl Linnaeus
(D) William Harvey

Q.4

The Ebony, Mahogany, Cinchona etc. are trees found in which type of forests in India?

(A) Tropical Deciduous Forests
(B) Tropical Evergreen Forests
(C) Thorny Forests
(D) Mangrove Forests

Q.5

In October 2023, Raghubar Das has been appointed the Governor of which of the following states of India?

(A) West Bengal
(B) Chhattisgarh
(C) Maharashtra
(D) Odisha

Q.6

What kind of structures are generally used by Buddhists for containing remains of their monks and nuns?

(A) Viharas
(B) Garbha griha
(C) Grihas
(D) Stupas

Q.8

Jaiminiya and Kausitaki dances are mentioned in which of the following?

(A) Puranas
(B) Brahmanas
(C) Agamas
(D) Vedas

Q.9

Geeta Chandran received the Padma Shri in 2007 who is an acclaimed _____ dancer.

(A) Bharatanatyam
(B) Sattriya
(C) Odissi
(D) Mohiniyattam

Q.11

Analgesic medicines are used for _____.

(A) correcting deficiencies due to vitamins and nutrients
(B) supporting physiological functions
(C) killing or weakening microbes and infections
(D) reducing pain and in fever

Q.12

Ustad Ghagge Khuda Baksh was associated with which of the following gharanas?

(A) Delhi gharana
(B) Agra gharana
(C) Gwalior gharana
(D) Benaras gharana

Q.13

Who was the first feudatory of Pallavas?

(A) Harshvardhan
(B) Vijayalaya
(C) Ashoka
(D) Rajendra

Q.14

As per the National Multidimensional Poverty Index Report 2021, which of the following states has the least population under poverty?

(A) Tamil Nadu
(B) Sikkim
(C) Himachal Pradesh
(D) Punjab

Q.15

Who among the following was the Viceroy of India during the announcement of the Partition of Bengal in 1905?

(A) Lord Dalhousie
(B) Lord Curzon
(C) Lord Canning
(D) Lord Hastings

Q.16

Which of the following is NOT an important function of the Supreme Court?

(A) It is the protector and guarantor of fundamental rights
(B) It maintains federal equilibrium
(C) It acts to check on executive authorities and enforce the rule of law
(D) It maintains of land records and revenue collection

Q.17

Which of the following is considered as a day of mourning among the Muslim community?

(A) Muharram
(B) Bakri Eid
(C) Shab-e-Barat
(D) Barah Wafat

Q.18
In which of the following states is shifting agriculture locally called Penda?

(A) Madhya Pradesh
(B) Orissa
(C) West Bengal
(D) Andhra Pradesh

Q.19
In January 2023, the Warehouse Development Regulation Authority (WDRA) signed a memorandum of understanding with _____ of India to help farmers get low interest rate loans.

(A) Bank of Baroda
(B) Canara Bank
(C) Union Bank of India
(D) State Bank of India

Q.20
What is the standard length of an Olympic-size swimming pool?

(A) 40 metres
(B) 60 metres
(C) 50 metres
(D) 70 metres

Q.21
Who among the following was the captain of the Indian cricket team during the 1983 Cricket World Cup event?

(A) Kapil Dev
(B) Sachin Tendulkar
(C) Mohinder Amarnath
(D) Sandeep Patil

Q.22
All living organisms on the Earth are made up of carbon compounds. Many years ago, dead and buried plants started slowly modifying into coal under high temperature and high pressure. This process is called:

(A) black gold formation
(B) peat formation
(C) carbonization
(D) composting

Q.23
Which Article of the Constitution of India lays down qualifications for the membership of the Parliament?

(A) Article 84
(B) Article 80
(C) Article 85
(D) Article 105

Q.24
Who among the following has written the famous novels (originally in Bangla) ‘Devdas’ and ‘Parineeta’?

(A) Rabindra Nath Tagore
(B) Sarat Chandra Chattopadhyay
(C) Bankim Chandra Chatterjee
(D) Nabin Chandra Sen

Q.25
Recreation services are an example of _____.

(A) capital goods
(B) consumer goods
(C) intermediate goods
(D) inferior goods

English Language and Comprehension 25 questions

Q.1
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. Maria was unhappy / because she terribly performed / in the examination / conducted last month.

(A) conducted last month
(B) because she terribly performed
(C) in the examination
(D) Maria was unhappy

Q.2

Select the most appropriate ANTONYM of the underlined word. Ritesh remarked, "Honesty is a great “virtue”."

(A) Probity
(B) Wickedness
(C) Integrity
(D) Rectitude

Q.3

Select the most appropriate option to fill in the blank and complete the expression. I begin to pronounce the sequence of words and numbers that will prevent her from giving him a piece of her _____.

(A) mind
(B) cake
(C) music
(D) art

Q.4

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don’t find any error, mark ‘No error’ as your answer. She was amazed by / the unique architecture of / the Sydney Opera House.

(A) the unique architecture of
(B) the Sydney Opera House
(C) She was amazed by
(D) No error

Q.5

Select the most appropriate option to fill in the blank. There was no one left in the world whom they could call _____.

(A) there's
(B) their's
(C) their
(D) theirs

Q.6

Select the grammatically correct sentence.

(A) It is the obvious that child needs love and attention of parents.
(B) It is obvious that a child needs love and attention of parents.
(C) It is a obvious that child needs love and attention of parents.
(D) It is an obvious that child needs love and attention of parents.

Q.7

Select the most appropriate ANTONYM of the word in brackets to fill in the blank. He spoke softly and _____ (tensely), trying to diffuse the tense situation.

(A) excitingly
(B) calmly
(C) madly
(D) furiously

Q.8

Select the most appropriate synonym of the given word. Exemplary

(A) Ordinary
(B) Average
(C) Excellent
(D) Bad

Q.9

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. He told / his senior, "I / will sung a song / at the party tonight."

(A) will sung a song
(B) He told
(C) at the party tonight."
(D) his senior, "I

Q.10

Select the most appropriate meaning of the given word. Improbable

(A) Tenable
(B) Implausible
(C) Plausible
(D) Practical

Q.11

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in the blanks. The touch-sensitive surface is one of the reasons why tablets and smart phones are so smart. All touchscreens generate an electric field _____ the screen. Your finger interferes with the field, and the device's operating system _____ your touch or swipe.

(A) beyond; escorts
(B) across; follows
(C) onto; chases
(D) past; attends

Q.12

Select the correctly spelt word to fill in the blank and make the sentence meaningful. Anuradha travelled to Italy last summer and _____ the historic ruins of Rome.

(A) explaured
(B) explored
(C) explord
(D) acsplored

Q.13

Select the option that can be used as a one-word substitute for the given group of words. A room in a hospital used especially for sunbathing or therapeutic exposure to light

(A) Solarium
(B) Planetarium
(C) Sanatorium
(D) Podium

Q.14

In the following sentence the underlined part contains as error. Alternatives to the underlined part are given as options. Select the correct alternative. It is very “hottest” over here.

(A) hot
(B) hotter
(C) most hot
(D) more hot

Q.15

Select the most appropriate synonym of the underlined word. He is “reluctant” to do the work.

(A) eager
(B) hesitant
(C) happy
(D) disposed

Q.16

Identify the INCORRECTLY spelt word.

(A) Deliverence
(B) Alliteration
(C) Banquet
(D) Condescend

Q.17

Select the most appropriate option to fill in the blank. The students were captivated by the _____ story told by their teacher.

(A) mundane
(B) superfluous
(C) compelling
(D) obsolete

Q.18

Correct the underlined part of the sentence. The rabbit runs “fastly”.

(A) fastest
(B) as fast as
(C) faster
(D) fast

Q.19

Select the most appropriate meaning of the given idiom. It's raining cats and dogs.

(A) There are cats and dogs in the rain.
(B) The rain is very light.
(C) It's a perfect day for a picnic.
(D) It's raining very heavily.

Q.20

Select the INCORRECTLY spelt word.

(A) Conscientous
(B) Necessary
(C) Committee
(D) Achieve

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As community-minded creatures who want (1)_____ to belong, we often mirror others, including on social media, where we adopt phrasing, tone and expressions of emotion. In the past few years, social media has had a (2)_____ on hyper positivity: think cheery emojis and (3)_____ quotes plastered over sunsets. Some put this 'good vibes only' trend down to the pandemic and a (4)_____ to avoid painful feelings when (5)_____ on difficult realities. However, attempting to convey constant happiness is not only difficult but impossible.

Select the most appropriate option to fill in blank number 1.

(A) laudatory
(B) deeply
(C) crisply
(D) abiotically

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As community-minded creatures who want (1)_____ to belong, we often mirror others, including on social media, where we adopt phrasing, tone and expressions of emotion. In the past few years, social media has had a (2)_____ on hyper positivity: think cheery emojis and (3)_____ quotes plastered over sunsets. Some put this 'good vibes only' trend down to the pandemic and a (4)_____ to avoid painful feelings when (5)_____ on difficult realities. However, attempting to convey constant happiness is not only difficult but impossible.

Select the most appropriate option to fill in blank number 2.

(A) dispersion
(B) disregard
(C) blemish
(D) focus

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As community-minded creatures who want (1)_____ to belong, we often mirror others, including on social media, where we adopt phrasing, tone and expressions of emotion. In the past few years, social media has had a (2)_____ on hyper positivity: think cheery emojis and (3)_____ quotes plastered over sunsets. Some put this 'good vibes only' trend down to the pandemic and a (4)_____ to avoid painful feelings when (5)_____ on difficult realities. However, attempting to convey constant happiness is not only difficult but impossible.

Select the most appropriate option to fill in blank number 3.

(A) reticent
(B) deterring
(C) motivational
(D) inobtrusive

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As community-minded creatures who want (1)_____ to belong, we often mirror others, including on social media, where we adopt phrasing, tone and expressions of emotion. In the past few years, social media has had a (2)_____ on hyper positivity: think cheery emojis and (3)_____ quotes plastered over sunsets. Some put this 'good vibes only' trend down to the pandemic and a (4)_____ to avoid painful feelings when (5)_____ on difficult realities. However, attempting to convey constant happiness is not only difficult but impossible.

Select the most appropriate option to fill in blank number 4.

(A) indolence
(B) listlessness
(C) desire
(D) apathy

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As community-minded creatures who want (1)_____ to belong, we often mirror others, including on social media, where we adopt phrasing, tone and expressions of emotion. In the past few years, social media has had a (2)_____ on hyper positivity: think cheery emojis and (3)_____ quotes plastered over sunsets. Some put this 'good vibes only' trend down to the pandemic and a (4)_____ to avoid painful feelings when (5)_____ on difficult realities. However, attempting to convey constant happiness is not only difficult but impossible.

Select the most appropriate option to fill in blank number 5.

(A) laminating
(B) ruminating
(C) stagnating
(D) collaborating

Answer key

Numerical and Mathematical Ability

Q.2 **(D)**	Q.3 **(C)**	Q.4 **(C)**	Q.5 **(D)**	Q.6 **(C)**	Q.7 **(B)**
Q.8 **(B)**	Q.9 **(B)**	Q.10 **(A)**	Q.12 **(C)**	Q.14 **(D)**	Q.15 **(D)**
Q.16 **(A)**	Q.18 **(D)**	Q.19 **(D)**			

Reasoning Ability and Problem Solving

Q.1 **(B)**	Q.2 **(C)**	Q.5 **(C)**	Q.6 **(A)**	Q.9 **(C)**	Q.10 **(A)**
Q.11 **(C)**	Q.12 **(C)**	Q.13 **(A)**	Q.15 **(A)**	Q.16 **(D)**	Q.17 **(B)**
Q.18 **(B)**	Q.19 **(B)**	Q.20 **(C)**			

General Awareness

Q.1 **(C)**	Q.3 **(A)**	Q.4 **(B)**	Q.5 **(D)**	Q.6 **(D)**	Q.8 **(B)**
Q.9 **(A)**	Q.11 **(D)**	Q.12 **(B)**	Q.13 **(B)**	Q.14 **(B)**	Q.15 **(B)**
Q.16 **(D)**	Q.17 **(A)**	Q.18 **(D)**	Q.19 **(D)**	Q.20 **(C)**	Q.21 **(A)**
Q.22 **(C)**	Q.23 **(A)**	Q.24 **(B)**	Q.25 **(B)**		

English Language and Comprehension

Q.1 **(B)**	Q.2 **(B)**	Q.3 **(A)**	Q.4 **(D)**	Q.5 **(D)**	Q.6 **(B)**
Q.7 **(B)**	Q.8 **(C)**	Q.9 **(A)**	Q.10 **(B)**	Q.11 **(B)**	Q.12 **(B)**
Q.13 **(A)**	Q.14 **(A)**	Q.15 **(B)**	Q.16 **(A)**	Q.17 **(C)**	Q.18 **(D)**
Q.19 **(D)**	Q.20 **(A)**	Q.21 **(B)**	Q.22 **(D)**	Q.23 **(C)**	Q.24 **(C)**
Q.25 **(B)**					

Explanations

Numerical and Mathematical Ability

Q.2 · Answer (D) 3 : 2

At 20% profit, ₹96 is 1.2 times the cost, so the mixture costs ₹80 a kg. By alligation, ghee : oil = (80 – 50) : (100 – 80) = 30 : 20 = 3 : 2.

Q.3 · Answer (C) 9 : 143

Slant height = $\sqrt{(16^2 + 12^2)}$ = 20 cm. The small cone on top has height 3 cm, radius 4 cm and slant 5 cm: TSA = π × 4 × (4 + 5) = 36π. The frustum has radii 16 and 4 cm and slant 15 cm: TSA = π(16 + 4) × 15 + π × 16^2 + π × 4^2 = 572π. 36 : 572 = 9 : 143.

Q.4 · Answer (C) 81.5

Weighted grade = 0.4 × 80 + 0.3 × 90 + 0.3 × 75 = 32 + 27 + 22.5 = 81.5.

Q.5 · Answer (D) 129

Total distance = 9 × 88 = 792 km. The first five hours cover 5 × 74 = 370 km and the last two 2 × 82 = 164 km, leaving 258 km for the sixth and seventh hours: 258 ÷ 2 = 129 km/hr.

Q.6 · Answer (C) 6.36

Cost price = 20 ÷ 1.1 = ₹18.18. For a 45% profit the price must be 18.18 × 1.45 = ₹26.36, an increase of ₹6.36.

Q.7 · Answer (B) 62.50

A single 30% discount on ₹5,000 is ₹1,500. The successive discounts leave 5,000 × 0.75 × 0.95 = ₹3,562.50 to pay, a discount of ₹1,437.50. The difference is ₹62.50.

Q.8 · Answer (B) 400

SI = 2,500 × 8 × 2/100 = ₹400.

Q.9 · Answer (B) 196 sq. ft

The largest square that fills the floor exactly has a side equal to the HCF of 98 and 70, which is 14 feet: 14^2 = 196 sq. ft.

Q.10 · Answer (A) 2.4 km

While the policeman runs 3 units the thief runs 2, so he gains 1 unit for every 3 he runs. To gain 800 m he runs 3 × 800 = 2,400 m = 2.4 km.

Q.12 · Answer (C) 6cm

The volume is unchanged, so $R^3 = 5^3 + 3^3 + 4^3$ = 125 + 27 + 64 = 216 and R = 6 cm.

Q.14 · Answer (D) 8

The five innings total 5 × 43 = 215 runs. The other four total 92 + 53 + 35 + 27 = 207, so X = 215 – 207 = 8.

Q.15 · Answer (D) 50.8

(40 + 49 + 56 + 72 + 37)/5 = 254/5 = 50.8 kg.

Q.16 · Answer (A) 50

All 50 together weigh 50 × 54 = 2,700 kg and the boys 20 × 60 = 1,200 kg, so the 30 girls weigh 1,500 kg: 1,500/30 = 50 kg.

Q.18 · Answer (D) 5 : 9

Expenditure = price × quantity: (5 × 7) : (7 × 9) = 35 : 63 = 5 : 9.

Q.19 · Answer (D) 600

SI = 2,000 × 6 × 5/100 = ₹600.

Reasoning Ability and Problem Solving

Q.1 · Answer (B) Both conclusions I and II follow
All parrots are eagles, so some eagles are parrots (I). The parrots that are ostriches are also eagles, so some eagles are ostriches (II). Both follow.

Q.2 · Answer (C) 24 and 54
Interchanging 24 and 54: (54 ÷ 6) + (18 × 4) – 24 + 42 = 9 + 72 – 24 + 42 = 99.

Q.5 · Answer (C) F
Going round the table to each person's right: F, D, B, E, C, A, G, and back to F (D is right of F, B right of D, E right of B, C right of E, A right of C, G right of A). So F sits to G's immediate right.

Q.6 · Answer (A) 3
T, R and F are in both words, as are 5, 6 and 9 in both codes. The letter left in RAFT is A and the digit left in its code is 3, so A = 3.

Q.9 · Answer (C) SZVR
Each letter moves 2 places forward: N → P, U → W, Q → S, M → O and J → L, Q → S, M → O, I → K. So QXTP → SZVR.

Q.10 · Answer (A) ZBE
Each letter moves 6 places forward from one cluster to the next, wrapping round after Z: B, H, N, T, Z; D, J, P, V, B; G, M, S, Y, E. The next cluster is ZBE.

Q.11 · Answer (C) 12
After interchanging the signs: 5 – 3 + 10 ÷ 2 × 2 = 5 – 3 + 10 = 12.

Q.12 · Answer (C) (4, 256, 4)
The second number is the first raised to the power of the third: $8^2 = 64$ and $12^1 = 12$. Likewise $4^4 = 256$.

Q.13 · Answer (A) 3
Number the seats 1 to 7 going to E's right, with E in seat 1. F is then in seat 4, C (two people from F) in seat 7 and B (three people from E) in seat 5. D is immediately to A's left, which leaves A in seat 3, D in seat 2 and G in seat 6. Counting to A's left come D, E and C, then G: three people.

Q.15 · Answer (A) 133
The differences alternate +2 and +6: 115, 117, 123, 125, 131, so the next term is 131 + 2 = 133.

Q.16 · Answer (D) 2
I, N and H are in both words, as are 9, 6 and 3 in both codes. The letter left in THIN is T and the digit left in its code is 2, so T = 2.

Q.17 · Answer (B) Only conclusion (II) follows.
The ants that are flies are bees, since all flies are bees, so some ants are bees (II). That contradicts I, which does not follow.

Q.18 · Answer (B) QHEZ
The letters move –1, –2, –1, –2: ODGM → NBFK and TAYP → SYXN. So RJFB → QHEZ (B moved back 2 wraps round to Z).

Q.19 · Answer (B) 123
The second number is six times the first plus 9: 12 × 6 + 9 = 81 and 24 × 6 + 9 = 153. Likewise 19 × 6 + 9 = 123.

Q.20 · Answer (C) ak
'hands' is in the first two sentences and 'du' in both their codes, so hands = du; likewise hearts = fd (sentences 2 and 3) and ears = bq (1 and 3). In the second sentence that leaves eyes = ak.

General Awareness

Q.1 · Answer (C) 2.25 Hz
Frequency is beats per second: 135 ÷ 60 = 2.25 Hz.

Q.3 · Answer (A) Rudolf Virchow
Rudolf Virchow (1855) proposed that new cells arise only from pre-existing cells: 'omnis cellula e cellula'.

Q.4 · Answer (B) Tropical Evergreen Forests
NCERT names ebony, mahogany, rosewood, rubber and cinchona as important trees of India's tropical evergreen forests.

Q.5 · Answer (D) Odisha
Raghubar Das, a former Chief Minister of Jharkhand, was appointed Governor of Odisha in October 2023.

Q.6 · Answer (D) Stupas
Stupas are mounds built over relics of the Buddha or of revered monks and nuns. Viharas were monasteries where monks lived, and the garbha griha is a temple's sanctum.

Q.8 · Answer (B) Brahmanas
SSC's key gives the Brahmanas: the Jaiminiya and the Kausitaki are Brahmanas, prose texts attached to the Sama Veda and the Rig Veda, and are said to mention dance together with music.

Q.9 · Answer (A) Bharatanatyam
Geeta Chandran, an acclaimed Bharatanatyam dancer, received the Padma Shri in 2007.

Q.11 · Answer (D) reducing pain and in fever
Analgesics reduce or abolish pain; many, such as aspirin and paracetamol, also bring down fever. Antimicrobials act against microbes, and supplements make up deficiencies.

Q.12 · Answer (B) Agra gharana
Ghagge Khuda Baksh is credited with bringing khayal singing into the Agra gharana and is often called its founder.

Q.13 · Answer (B) Vijayalaya
SSC's key gives Vijayalaya, the only one of the four linked to the Pallavas: he is generally held to have been their feudatory when he took Thanjavur in the mid-ninth century and founded the imperial Chola line. Harshavardhana and Ashoka ruled in the north, and Rajendra was a later Chola emperor.

Q.14 · Answer (B) Sikkim
In NITI Aayog's National Multidimensional Poverty Index 2021 (based on NFHS-4, 2015-16), Sikkim's share of poor people (3.82%) was lower than Tamil Nadu's (4.89%), Punjab's (5.59%) and Himachal Pradesh's (7.62%).

Q.15 · Answer (B) Lord Curzon
Lord Curzon was Viceroy when the Partition of Bengal was announced in 1905; it took effect on 16 October 1905.

Q.16 · Answer (D) It maintains of land records and revenue collection
Land records and revenue collection are the work of state revenue departments. The Supreme Court guards Fundamental Rights, settles disputes between the Union and the states, and checks the executive through judicial review.

Q.17 · Answer (A) Muharram
Muharram, above all its tenth day, Ashura, is observed as a time of mourning for Imam Husain, killed at Karbala in 680 AD.

Q.18 · Answer (D) Andhra Pradesh
NCERT lists the local names of slash-and-burn (shifting) agriculture: 'Podu' or 'Penda' in Andhra Pradesh, 'Bewar' or 'Dahiya' in Madhya Pradesh, and 'Pama Dabi', 'Koman' or 'Bringa' in Odisha.

Q.19 · Answer (D) State Bank of India
In January 2023 the WDRA and the State Bank of India signed an MoU to promote SBI's Produce Marketing Loan, lent to farmers against electronic negotiable warehouse receipts (e-NWRs) at low interest.

Q.20 · Answer (C) 50 metres
An Olympic-size swimming pool is 50 metres long.

Q.21 · Answer (A) Kapil Dev
Kapil Dev captained India to its first World Cup title in 1983, beating the West Indies in the final at Lord's.

Q.22 · Answer (C) carbonization
The slow conversion of buried dead vegetation into coal under high temperature and pressure is called carbonisation.

Q.23 · Answer (A) Article 84
Article 84 sets out the qualifications for membership of Parliament. Article 80 deals with the composition of the Rajya Sabha, Article 85 with the sessions of Parliament and Article 105 with members' privileges.

Q.24 · Answer (B) Sarat Chandra Chattopadhyay
Sarat Chandra Chattopadhyay wrote both 'Devdas' and 'Parineeta'.

Q.25 · Answer (B) consumer goods
Recreation services are bought by households for their own final use, so they count as consumer goods (consumer services). Capital and intermediate goods are used in further production, and 'inferior' goods are those whose demand falls as income rises.

English Language and Comprehension

Q.1 · Answer (B) because she terribly performed
The adverb belongs after the verb: 'because she performed terribly'.

Q.2 · Answer (B) Wickedness
'Virtue' is moral goodness; its opposite is 'wickedness'. 'Probity', 'integrity' and 'rectitude' are synonyms.

Q.3 · Answer (A) mind
To give someone 'a piece of one's mind' is to tell them angrily what one thinks of them.

Q.4 · Answer (D) No error
The sentence is correct: 'amazed by' takes what causes the amazement, here 'the unique architecture of the Sydney Opera House'.

Q.5 · Answer (D) theirs
The possessive pronoun 'theirs' stands without a noun: someone they could call their own. 'Their' needs a noun after it, 'there's' means 'there is', and 'their's' is not a word.

Q.6 · Answer (B) It is obvious that a child needs love and attention of parents.
The predicative adjective 'obvious' takes no article ('It is obvious that'), while the singular countable noun 'child' needs one: 'a child needs'.

Q.7 · Answer (B) calmly
The opposite of 'tensely' is 'calmly', which fits speaking softly to ease a tense situation.

Q.8 · Answer (C) Excellent
'Exemplary' means good enough to serve as an example: 'excellent'.

Q.9 · Answer (A) will sung a song
'Will' takes the base form of the verb: 'I will sing a song'. 'Sung' is the past participle.

Q.10 · Answer (B) Implausible
'Improbable' means unlikely to be true or to happen: 'implausible'. 'Plausible' is its opposite.

Q.11 · Answer (B) across; follows
The field spreads 'across' the screen, and the operating system 'follows', that is tracks, the touch or swipe. 'Escorts', 'chases' and 'attends' do not describe how a device registers a touch.

Q.12 · Answer (B) explored
The correct spelling is 'explored'.

Q.13 · Answer (A) Solarium
A solarium is a room built to let in sunlight, for sunbathing or light therapy. A sanatorium treats long-term illness, a planetarium shows the night sky, and a podium is a raised platform.

Q.14 · Answer (A) hot
'Very' goes with the positive degree, not the superlative: 'It is very hot over here.'

Q.15 · Answer (B) hesitant
'Reluctant' means unwilling: 'hesitant'. 'Eager' and 'disposed' are near opposites.

Q.16 · Answer (A) Deliverence
The correct spelling is 'deliverance'.

Q.17 · Answer (C) compelling
A story that captivated the students was 'compelling', one that holds attention. 'Mundane' (dull), 'superfluous' (unnecessary) and 'obsolete' (outdated) do not fit.

Q.18 · Answer (D) fast
'Fast' is both an adjective and an adverb, and 'fastly' is not standard English: 'The rabbit runs fast.'

Q.19 · Answer (D) It's raining very heavily.
'It's raining cats and dogs' means it is raining very heavily.

Q.20 · Answer (A) Conscientous
The correct spelling is 'conscientious'.

Q.21 · Answer (B) deeply
Creatures who want 'deeply' to belong: the adverb fits the strong human need to belong.

Q.22 · Answer (D) focus
Social media has had a 'focus' on hyper positivity, as the cheery emojis and upbeat quotes that follow show.

Q.23 · Answer (C) motivational
'Motivational' quotes plastered over sunsets, like cheery emojis, are examples of the hyper-positive trend.

Q.24 · Answer (C) desire
A 'desire' to avoid painful feelings; 'indolence', 'listlessness' and 'apathy' all mean a lack of energy or interest.

Q.25 · Answer (B) ruminating
'Ruminating on' difficult realities is thinking about them over and over.

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